INCH-POUND

MIL-P-60596B (AR)
30 April 1990
SUPERSEDING
MIL-P-60596A (AR)
25 March 1977

MILITARY SPECIFICATION

PROJECTILE, HE, 8 INCH M106 METAL PARTS ASSEMBLY

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Projectile, HE, 8 Inch, M106, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning method and pretreatment of ferrous surfaces for organic coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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MILITARY

MIL-H-6875	- Heat Treatment of Steel, Process For
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-S-10520	- Steel, Forging for Projectile Stock
MIL-S-11310	- Steel Bars, Carbon, Hot Rolled, for Cold Shaping, Including Cold Extrusion

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Table for Inspection by Attributes
MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1168	- Lot Numbering of Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9341621	- Packing and Marking for Pallet, Wood, Ammunition for Projectile, 8 Inch, HE, M106
10534909	- Projectile, 8 Inch, HE, M106 Metal Parts Assembly

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INSPECTION EQUIPMENT DRAWINGS

8593429	- Band Seating Testing Machine Indentor and Test Data
9225691	- Ultrasonic Inspection Standard, Longitudinal and Radial Defects 8 Inch M106 Projectile Body
9225692	- Ultrasonic Inspection Standard, Circumferential and Piping Defects 8 Inch, M106 Projectile Body
9225694	- Envelope Drawing, Ultrasonic Inspection Equipment, M106 Projectile Body

PUBLICATIONS

Frankford Arsenal
Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection

(Copies of specificaitons, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 711	- Carbon Steel Blooms, Billets and Slabs for Forging
ASTM B 117	- Salt Spray (Fog) Testing
ASTM E 8	- Tension Testing of Metallic Materials
ASTM E 10	- Brinell Hardness of Metallic Materials
ASTM E 103	- Rapid Indentation Hardness Testing of Metallic Materials

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Steel quality. The item contractor shall establish procedures to provide a forge quality steel when procuring steel in accordance with ASTM A 711, i.e., the steel shall be free of pipe, porosity and heavy segregation. The manufacturer will provide a certification to cite the specific revisions of the industry standard under which the steel was manufactured. Material procured in accordance with MIL-S-10520 or MIL-S-11310 shall comply with the requirements of the applicable specification.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing. Furnace equipment requirement and furnace calibration and test procedures shall be in accordance with the requirements of MIL-H-6875.

3.3.1 Hot forge - heat treat method. Prior to finish machining projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (F), followed by an oil quench. After quenching, the body forgings shall be tempered without delay at a temperature above 750 degrees F suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees F, after which cooling may be accelerated by water spray.

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3.3.2 Cold shaping and hot cup - cold draw methods. Cold working and subsequent stress relief shall be accomplished as to produce in the projectile body the mechanical properties specified by the applicable drawing. Stress relief shall be accomplished at a temperature of 900 degrees F minimum following the final forming operation which results in plastic flow. The minimum time the projectile body shall be at the minimum stress relief temperature shall be one hour.

3.3.2.1 Process control of stress relief operation. A process control document covering the stress relief operation shall be submitted to the technical agency through the contracting officer for review and approval prior to the initiation of production. This document shall include descriptions of procedures, equipment, process parameters, monitoring practices and calibration procedures and frequencies to be used for the stress relief operation. Data shall be included showing, based on temperature measurements of both the nose and base area of the projectile, that the projectile bodies are a minimum temperature for the minimum time in accordance with 3.3.2 when using the process parameters outlined in the process control document. The process control document shall be a part of the contractor's quality program plan. Revisions shall be submitted to the technical agency through the contracting officer for review and approval.

3.4 Mechanical properties.

3.4.1 Body. Mechanical properties specified on the applicable drawing shall apply throughout the component.

3.4.2 Rotating band. After application and subsequent removal, the rotating band shall have the following mechanical properties:

Tensile Strength - 30,000 lbs. per square inch (psi) minimum
Elongation - 25 percent per inch of gage length, minimum (min.)

3.5 Banding.

3.5.1 Band seat. Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.5.2 Band blank. (Gilding Metal) The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification, except that where hot banding at not less than 800 degrees Fahrenheit (°F). No prior annealing is necessary.

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3.5.3 Application. The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the sidewall depth. The band shall be applied in such a manner as not to distort the body wall. There shall be no visible gap between the rotating band and the sidewalls of the band seat (on the outside surface of the projectile) after the rotating band has been machined to final dimensions.

3.5.3.1 Application hot or cold. The bands may be applied either hot or cold but if applied hot, the bands shall not be heated above 1600°F to prepare them for application to the body. The bands shall be heated uniformly be a process which will prevent undue oxidation (loose scale), and applied in a manner that will hold the formation of scale between the band and the band seat to a minimum.

3.5.4 Band seating. The indicated diametrical clearance between the rotating band and band seat shall not exceed .025 inch, when non-destructive method is used.

3.6 Metal defects.

3.6.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.6.2 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Visual standards. Visual standards for metal defects of paragraph 3.6.2 will be established for each contract after the first months production, or whenever requested by the procuring activity (see 6.8).

3.7 Surface finish. The requirements for surface finish are as detailed on the applicable drawing.

3.8 Ultrasonic inspection. The projectile shall be free of any discontinuities equal to or in excess of the inspection criteria within their respective zones. The inspection criteria are notches and holes in standards (dwgs. 9225691, 9225692) which simulate rejectable cracks and piping flaws (that is, all projectiles shall have an echo signal that is less than the signal obtained from the standards).

3.9 Proving ground functional test. Projectiles, when tested at excess pressure, shall show no evidence of loss or break up of metal parts components in flight.

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3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.11.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.11.2 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.11.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.11.4 Base cover weld. The weld shall be sound, and the welding operation shall be so controlled that there will be adequate fusion between projectile base and base cover.

3.11.5 Burr. No part shall have a burr which might interfere with the assembly of the round or which might be injurious to personnel handling the item.

3.11.6 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter, prior to inspection and shipping.

3.11.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred and otherwise damaged in subsequent handling prior to shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Cover, Base	10534912	5
Blank, Rotating Band	10522521	5
Body, (Prior to to band seat machining and threading)	10534911	5
Body, (Prior to nosing)	10534911	5
Body	10534911	5
Body, Assembly	10534910	5
Projectile, Metal Parts Assembly	10534909	55

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60596B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER See below
	Projectile Assemblies and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cover, Base				
	(Dwg. 10534912)				
	Examination for defects	5	100%	3.1	4.4.2.1
	Blank, Rotating Band				
	(Dwg. 10522521)				
	Mechanical properties	5	100%	3.5.2	4.5.2.2.2
	Body (prior to nosing)				
	(Dwg. 10534911)	5	100%	3.6	4.4.2.2/4.5.1
	Body, (prior to band seat machining) and threading nose				
	(Dwg. 10534911)				
	Examination for defects	5	100%	3.1	4.4.2.3
	Ultrasonic test	5	100%	3.8	4.5.4
	Body				
	(Dwg. 10534911)				
	Examination for defects	5	100%	3.1	4.4.2.4
	Mechanical properties				
	Tensile strength	2	100%	3.4	4.5.2.2.1

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-60596B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile Assemblies and Components				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body assembly				
	(Dwg. 10534910)				
	Examination for defects	5	100%	3.1	4.4.2.5
	Band seating	5	100%	3.5.4	4.5.3
	Mechanical properties-rotating band	5	100%	3.5.2	4.5.2.2.3
	Base cover weld test	5	100%	3.11.4	4.5.5
	Projectile Metal Parts Assembly				
	(Dwg. 10534909)				
	Examination for defects	5	100%	3.1	4.4.2.6
	Salt spray testing	5	100%	3.1	4.5.6.1
	Protective coating thickness	5	100%	3.1	4.5.6.2
	Functional test (proving ground)				
	(Dwg. 10534909)				
	Ballistic testing	50	100%	3.9	4.5.7.1

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, the following applies:

a. Heat treatment lots. Heat treatment lots shall contain heat treated bodies from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

b. Stress relieved lots. Stress relieved lots shall contain stress relieved bodies, from not more than one heat of steel, that have been cold shaped by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

c. Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lots code number and processed in accordance with either (a) or (b) above, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

d. Projectile lots. When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

e. Shipping lots. Shipping lots shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawings, revisions, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lots thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10534912
4.4.2.1	Base, Cover				NEXT HIGHER ASSEMBLY 10534910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.1	Gage
Minor					
201	Diameter		0.65%	3.1	Gage
202	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10534911
4.4.2.2	Body (prior to nosing)				NEXT HIGHER ASSEMBLY 10534910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special 1	Metal defective cavity		100%	3.6	4.5.1/Visual exam
Major	None defined				
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60596B (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Body (prior to band seat machining and threading nose)	1	1	10534911
				NEXT HIGHER ASSEMBLY: 10534910

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Ultrasonic test		100%	3.8	4.5.4
Special 1	Ultrasonic test stamp missing		100%	3.1	Visual
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 10534911
4.4.2.4	Body				NEXT HIGHER ASSEMBLY 1053491Ø
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
1Ø1	Circular runout of pitch diameter of fuze thread with body & front face		Ø.4Ø%	3.1	Gage
1Ø2	Circular runout of ogive with body		Ø.4Ø%	3.1	Gage
1Ø3	Circular runout of bourrelet with body		Ø.4Ø%	3.1	Gage
1Ø4	Circular runout of boattail with body		Ø.4Ø%	3.1	Gage
1Ø5	Location of basic diameter on ogive		Ø.4Ø%	3.1	Gage
1Ø6	Diameter of bourrelet, min.		Ø.4Ø%	3.1	Gage
1Ø7	Diameter of body		Ø.4Ø%	3.1	Gage
1Ø8	Wall thickness, min.		Ø.4Ø%	3.1	Gage
1Ø9	Wall thickness variation, max.		Ø.4Ø%	3.1	Gage
11Ø	Base thickness		Ø.4Ø%	3.1	Gage
111	Diameter of rotating band seat		Ø.4Ø%	3.1	Gage
112	Rotating band seat rib missing		Ø.4Ø%	3.1	Visual
113	Knurling or chisel cuts in band seat ribs missing		Ø.4Ø%	3.1	Visual
114	Undercut of band seat sidewall missing		Ø.4Ø%	3.1	Viaual
115	Foreign matter on band seat		Ø.4Ø%	3.11	Visual
116	Pitch diameter of fuze thread, max.		Ø.4Ø%	3.1	Gage
117	Minor diameter of fuze thread, max.		Ø.4Ø%	3.1	Gage
118	Length of effective thread, min.		Ø.4Ø%	3.1	Gage
119	Perpendicularity of rear face with body		Ø.4Ø%	3.1	Gage
12Ø	Mechanical properties test tension		(a)	3.4.1	4.5.2.2.1

NOTES:

(a) See paragraph 4.4.3.2.2.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	Body				10534911
					NEXT HIGHER ASSEMBLY
					10534910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Pitch diameter of fuze thread, min.		0.65%	3.1	Gage
202	length of thread flat, max.		0.65%	3.1	Gage
203	Location of boattail angle with respect to band seat		0.65%	3.1	Gage
204	Location of rotating band seat		0.65%	3.1	Gage
205	Width of rotating band seat		0.65%	3.1	Gage
206	Evidence of shot or sandblasting on band seat		0.65%	3.5.1	Visual
207	Diameter at nose		0.65%	3.1	Gage
208	Diameter at nose thread chamfer		0.65%	3.1	Gage
209	Location of bourrelet		0.65%	3.1	Gage
210	Length total		0.65%	3.1	Gage
211	Boattail angle		0.65%	3.1	Gage
212	Profile of ogive		0.65%	3.1	Gage
213	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (see 6.8)
214	Surface finish improper		0.65%	3.7	Visual/ANSI-B46.1
215	Burr		0.65%	3.11.5	Visual
216	Foreign matter		0.65%	3.11.6	Visual
217	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.5	Body Assembly (prior to painting)		10534910
			NEXT HIGHER ASSEMBLY
			10534909

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Metal defects (after phosphating)		100%	3.6	Visual
2	Missing base plate, broken base plate weld or perforated base plate		100%	3.1	Visual
Major					
101	Circular runout of rotating band flat with body		0.40%	3.1	Gage
102	Diameter of rotating band obturating lip		0.40%	3.1	Gage
103	Diameter of rotating band forward obturating lip		0.40%	3.1	Gage
104	Diameter of rotating band to rear of obturating lip		0.40%	3.1	Gage
105	Weight		0.40%	3.1	Balance scale
Minor					
201	Width of rotating band		0.65%	3.1	Gage
202	Location of rotating band obturating lip		0.65%	3.1	Gage
203	Location of rotating band forward lead angle		0.65%	3.1	Gage
204	Location of rear taper on rotating band		0.65%	3.1	Gage
205	Width of rotating band relief groove		0.65%	3.1	Gage
206	Diameter of rotating band connelures		0.65%	3.1	Gage
207	Diameter of rotating band relief groove		0.65%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.5	Body Assembly (prior to painting)		10534910
			NEXT HIGHER ASSEMBLY 10534909

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208	Rotating band relief groove angle		0.65%	3.1	Gage
209	Location of rotating band		0.65%	3.1	Gage
210	Position of base cover		0.65%	3.1	Gage
211	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (see 6.8)
212	Surface finish improper		0.65%	3.7	Visual/ANSI-B46.1
213	Burr		0.65%	3.11.5	Visual
214	Foreign matter		0.65%	3.11.6	Visual
215	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.6	Projectile, 8 Inch, HE, M106 Metal Parts Assembly (after painting)				10534909
					NEXT HIGHER ASSEMBLY 9207909
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Pool of paint in cavity		100%	3.1	Visual
2	Metal defects in cavity		100%	3.6	Visual
Major					
101	Diameter of bourrelet, max. (after painting)		100%	3.1	Gage
102	Rotating band damaged		0.40%	3.1	Visual
103	Foreign matter in cavity		0.40%	3.1	Visual
104	Protective coating thickness		(a)	3.1	4.5.6.2
105	Salt spray test		(a)	3.1	4.5.6.1
Minor					
201	Paint on threads		0.65%	3.1	Visual
202	paint on rotating band bearing surface		0.65%	3.1	Visual
203	Protective coating in cavity inadequate or damaged		0.65%	3.1	Visual
204	Protective coating on exterior surfaces inadequate or damaged		0.65%	3.1	Visual
205	Anti-seize compound missing from threads		0.65%	3.1	Visual
206	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

(a) See paragraph 4.4.3.5.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.6	Projectile, 8 Inch, HE, M106 Metal Parts Assembly (after painting)		10534909
			NEXT HIGHER ASSEMBLY 9207909

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Grommet missing		0.65%	3.1	Visual
208	Plug, shock attenuating missing		0.65%	3.1	Visual
209	Total length max.		0.65%	3.1	Gage
210	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Projectile, 8 Inch, HE, M106 Metal Parts Assy Destructive Lot Acceptance				10534909
					NEXT HIGHER ASSEMBLY 9207909
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Government Proving Ground Ballistic test (excess pressure)				
Critical 1	Components break-up or become disassembled in bore or in flight	(a)	100%	3.9	4.5.7.1
Major	None defined				
Minor	None defined				

NOTES:

(a) See paragraphs 4.4.4.1 and 4.4.4.2.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packing and Marking for Pallet, Wood Ammunition for Projectile, 8 Inch				9341621
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pallet damaged (unserviceable)		Ø.65%	3.1	Visual
202	Improperly packed		Ø.65%	3.1	Visual
203	Strapping missing, loose or broken		Ø.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		Ø.65%	3.1	Visual
205	Evidence of poor workmanship		Ø.65%	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing.

4.4.3.1 Steel quality. Records of test documentation shall be maintained and made available for Government review to provide evidence of compliance with the requirements of 3.2

4.4.3.2 Mechanical properties.

4.4.3.2.1 Hardness.

4.4.3.2.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.4.1a) after heat treatment, but prior to final machining of the bourrelet diameter, shall be subjected to a hardness test with a Brinell Hardness Tester or any other method approved by the procuring activity. Hardness impressions shall be made on the body bourrelet diameter and used as the basis for selecting the tension test samples of paragraph 4.4.3.2.2.1.

4.4.3.2.1.2 Stress relieved lots. Twenty-five (25) cold shaped bodies from the first batch of each stress relieved lot (as defined by 4.4.1b) processed through the stress relief furnace and 5 bodies from each one hours' production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product.

4.4.3.2.1.3 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.2.

4.4.3.2.2 Tension test.

4.4.3.2.2.1 Heat treatment lots. Two (2) projectile bodies, one of the hardest and one of the softest as determined by the hardness test, shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

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4.4.3.2.2.2 Stress relieved lots. Two (2) projectile bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five projectiles, shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five degrees Fahrenheit but not below the minimum temperature of 900 degrees Fahrenheit for the minimum one hour soak time. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each eight (8) hours by an independent method.

4.4.3.2.2.3 Miscellaneous lots. Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.4.3.2.2.1 or 4.4.3.2.2.2, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawings shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.4.3.2.2.4 Rotating band. One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subjected to this test to determine compliance with 3.4.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3 Band seating.

4.4.3.3.1 Tightness.

4.4.3.3.1.1 Nondestructive method. Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.5.4 shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3.1.2 Destructive method. When the nondestructive equipment is not available, one body assembly from each 8-hour production or fraction thereof from each banding machine in operation shall be subjected to this test to determine compliance with 3.5.4. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

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4.4.3.3.2 Sidewall contact. Rotating bands removed as specified under 4.4.3.2.2.4 shall be examined to determine compliance with 3.5.3. Failure to comply shall be cause for rejection of assemblies represented by the sample.

4.4.3.4 Base cover weld. Two body assemblies from each 1-hour production or fraction thereof per welding machine in operation shall be subjected to this test. Failure of the weld to meet requirements of 3.11.5 shall be cause for rejection of the assemblies represented by the sample.

4.4.3.5 Protective coating - salt spray. The sample size and acceptance criteria shall be as specified in the applicable specification. If failure occurs, all items processed since last acceptance shall be rejected and corrective action taken.

4.4.3.5.1 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.4 Proving ground functional tests. The following proving ground tests will be performed by the Government:

4.4.4.1 First article and initial production lot samples. Beginning with the first article produced and continuing until three consecutive production lots have complied with the applicable specified requirements, 50 projectiles shall be randomly selected from each lot for the excess pressure test. If any projectile fails to comply with the applicable requirements, the lot shall be rejected.

4.4.4.2 Production lot samples. After three consecutive initial production lots have met the criteria of 4.4.4.1, 5 projectiles shall be randomly selected from each lot for excess pressure test. If any projectile fails to comply with the applicable requirements, the lot shall be rejected.

4.4.5 Tool control dimensions.

a. Dimensions marked 'tool control' shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted providing the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

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b. When the contractor has satisfied (a) above, visual inspection of the tool control dimension(s) will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.4.6 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the table starting in Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Visual examination. The internal cavity of the projectile shall be inspected visually prior to nosing using peripheral lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.6.3 shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements of 3.6 shall be classed defective and removed from the lot.

4.5.2 Mechanical properties.

4.5.2.1 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone.

b. Hardness tester periodic verification during each operating shift shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The Hardness Tester shall use a standard 10 MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The

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preload shall not exceed 10% of the major load of 300Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

4.5.2.2 Tension test.

4.5.2.2.1 Tension test methods. From the samples provided by the heat treat, stress relief or miscellaneous lots two tension test specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by ASTM Method E-8, shall be taken longitudinally from the opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) percent.

4.5.2.2.2 Mechanical properties of rotating band blanks. The band blank shall be cut from front to rear and flattened, without hammering which may further cold work that material, and the largest obtainable flat type specimen shall be cut from the band, the length of which shall provide a one inch gage length, plus adequate material at each end to accommodate the grips of the tensile testing machine. This method should yield a specimen with a 0.500 inch gage width. In any event, the gage cross section shall be determined to the nearest 0.001 square inch and utilized to calculate the tensile and yield strengths, and elongation.

4.5.2.2.3 Mechanical properties of disassembled rotating band. The sample provided by 4.4.3.2.2.4 shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having one inch gage length and a one-half inch gage width, and grip ends of sufficient length to accommodate the tensile machine shall be removed from the blank and tested in accordance with the procedure prescribed by ASTM Method E-8.

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4.5.3 Band seating.

4.5.3.1 Tightness.

4.5.3.1.1 Non-destructive method. The samples provided by 4.4.3.3.1.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 6.10).

4.5.3.1.2 Destructive method. The body assembly selected in accordance with 4.4.3.3.1.2 should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.3.2 Sidewall contact. the samples provided in accordance with 4.4.3.2.2.4 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.5.3.

4.5.4 Ultrasonics. 100 percent inspection.

4.5.4.1 Inspection. The projectile shall be ultrasonically inspected, prior to cutting band seat and threads and application of base plate. Any projectile failing to meet the applicable requirements shall be classed defective and removed from the lot. Stamp the letter "U", as per the applicable drawing on each projectile accepted by ultrasonic inspection.

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4.5.4.2 Equipment and procedures. The equipment and process for ultrasonic inspection shall be in accordance with drawing 9225694.

4.5.4.3 Ultrasonic standards. The ultrasonic inspection standards shall be provided by the contractor for use during the ultrasonic inspection. These standards shall be machined from acceptable projectiles in accordance with drawings 9225691 and 9225692 and shall be inspected for all drawing requirements to determine acceptability for use. A certificate of conformance for the standards shall be submitted to the technical agency (see 6.5), prior to use of the standards.

4.5.4.4 Operator qualifications. For operation of ultrasonics equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II in all respects but need not include experience.

4.5.5 Base cover weld. The test of the base cover weld shall be performed as follows: Visually inspect the base cover and select the area that appears to have the weakest weld; insert a 5/8" chisel in this area and apply force by use of a one pound hammer and an approximate one foot stroke. If a weld breaks loose after 2 strokes, the sample will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked into position and accepted.

4.5.6 protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon (see 6.11).

4.5.6.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectile on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by ASTM Method B 117 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.5.1 and test samples shall be dried in accordance with Drawing 10534909.

4.5.6.2 Protective coating - thickness. Protective coating thickness shall be accomplished using commercially available equipment.

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b. the supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade selected by the projectile manufacturer and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

c. In the application of carbon steel, blooms, billets, slabs, and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions in weight per unit length, in the hardenability variance from heat to heat in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques have capabilities of compensating for such variables.

6.13 Heat treatment. Heat treatment is understood to be any thermal process which brings about a modification in the machanical and metallurgical properties of the metal involved.

6.14 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DOD FAR Supplement Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specificaiton are cited in the following paragraphs:

Paragraph	Data Requirement Title	Applicable DID
3.3.2.1	Process Control Document	DI-R-5297A

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6.15 Subject term (key word) listing.

Ammunition
Ballistic testing parts security
Military Specifications
Projectile; 8 Inch, M106
Rotating Band Seating (band seating testing machine)

6.16 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-AB11)

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4.5.7 Proving ground functional test.

4.5.7.1 Excess pressure. The ballistic test projectiles inert loaded and dummy fuzed will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. The propellant charge shall be adjusted to attain a pressure of between 44,300 to 48,100 psi. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressure will be recorded.

5. PACKAGING

5.1 Packing. Level C. Minimum military pack.

5.1.1 Packing. Projectile metal parts assemblies including lifting plug and grommet shall be palletized in accordance with drawing 9341621.

5.1.2 Marking. Marking shall be in accordance with drawing 9341621.

6. NOTES (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the Projectile 8 Inch, HE, M106.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection procedures and equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Research and Development command (ARRADCOM) or US Army Armement Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.5 Ultrasonic standards. The certificate of conformance for the ultrasonic standards shall be submitted to: Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, New Jersey 07806-5000, prior to use of standards.

6.6 Submission of test data and ammunition data cards. In addition to the normal distribution of records, when the projectile is procured by the US Army Armament Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: SMCAR-CCH-P and AMSMC-QAR-Q(D), Picatinny Arsenal, NJ 07806-5000, and Commander, ARRCOM, ATTN: AMSMC-QAD, Rock Island, IL 61299-6000.

6.7 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.8 Visual standards. Visual standards to be established at contractor's plant in compliance with 3.6.3.

6.9 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to, or better than, the quality assurance provisions specified herein.

6.10 Band seating. The operating procedures for use of Detroit Testing machine as prepared by the Inspection Equipment Branch, US Army Armament, Research, Development and Engineering Center, AMSMC-QAR-I(D), Picatinny Arsenal, New Jersey 07806-5000.

6.11 Protective coating thickness. In checking the thickness of protective coating the "Dermitron" by Unit Process Assemblies Inc., 53-15 37th Avenue, Woodside, New York 11377, has been proven to be satisfactory.

6.12 Steel quality.

a. It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions and mechanical properties and metal integrity specified as an end item requirement.

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